

GRATIFOOD

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ABSTRACT

We often take things for granted. One such thing is food. A lot of effort goes on in bringing the plate of food to our tables, and we recognize that effort when we cook for ourselves, but often we neglect that when we eat at restaurants. Invisible Workers (People working behind the scenes) ranging from farmers, people from production, distribution, storing, stocking, and many more to the chefs that cook our food are some of the people that have put a lot of effort into the food on our plates. The goal of this paper is to acknowledge and make the effort of invisible workers visible by raising awareness and offering a means to express gratitude. As a specific case, we decided to focus on workers at Indiana University Diners that represent a part of a huge population of invisible workers. Our design uses a Kiosk based system that will allow students who eat at Indiana university diner to share gratitude that will reach the invisible workers via diverse ways such as email, audio, etc.

CCS CONCEPTS

• **Human-Centered computing** → Human Computer Interaction (HCI); Empirical studies in HCI.

KEYWORDS

Gratitude, Invisible Workers, Dining area, University Students, Appreciation

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1 INTRODUCTION

Food is a very exclusive topic when it comes to showing gratitude. The emotion of showing gratitude in any household starts generally on the dinner table for the food and most of the time to people that we know. Appreciating food is also an important concept in mindful eating [4]. It is a major factor, especially in the aspect of children when they grow up. It has been found that the association between their appreciation for food and outcomes such as attitude towards food, eating behaviors, and health status [3]. Many non-human species such as primates and monkeys show gratitude to each other by sharing a kiss, hug, or food in response to sharing food. Many religions consider gratitude as the most important virtue or emotion in a person. We humans generally show appreciation or gratitude to

either ourselves in a personal general or to our family and friends by saying thank you or doing something nice for them. But we often do not show the same kind of gratitude to people that we don't know, or even for things that we use.

When we go out to a restaurant, we thank the waiter for serving the food in the form of a tip and a verbal 'Thank you' and sometimes even compliment the chef. All that appreciation and acknowledgment for food and the service goes to the person that is visible either on the counter or while serving food. But what about other employees working for the restaurant? What about people who get raw materials to the kitchen from the market? What about the farmers who produce food? Also, what about material things like water, soil? If we start tracing back to the source of things just for the food that we eat, it will be a never-ending list of people involved in bringing the food to the plate behind the scenes. In this paper, we use the term "Invisible workers" to refer to individuals who work behind the scenes.

Our initial research revealed that there are many invisible workers who are dissatisfied with their job as it lacks motivation or acknowledgment for their effort other than the wages. A review of relevant literature confirmed that gratitude has many individual benefits such as physical and mental health, perhaps it also inspires social behavior such as better relationships with others and improved workplace climate, motivation, and job satisfaction [10].

Based on these findings we decided to design a system called Gratifood that would make people eating at diners more aware of the invisible workers involved in the food that they are consuming and enable them a means to show their gratitude. Eventually, conveying this appreciation to invisible workers can possibly help them improve their mental health and feel motivated at work. As a starting point for this design solution, we decided to consider Indiana University (IU) Diners as a huge population of students visit IU Diners daily to eat throughout the day and this might even help us to reach out to many invisible workers involved in the food that is served at IU Diners.

We envision Gratifood as a powerful tool for diners and restaurants to raise awareness about food, the effort that goes into food and encourages sharing gratitude that can help invisible workers.

2 PRELIMINARY STUDY

We conducted preliminary research by taking surveys and semi-interviews to investigate both the notion of people working at dining and to understand whether the employed needed any form of visibility and later the effect of gratitude in their daily tasks at work. We wanted to understand their daily pattern of work at dining and their interaction with students, their awareness, and their attitude towards their work. We then synthesized and analyzed data and developed insights to inform our proposed design. Along with that we also wanted to understand how they show gratitude

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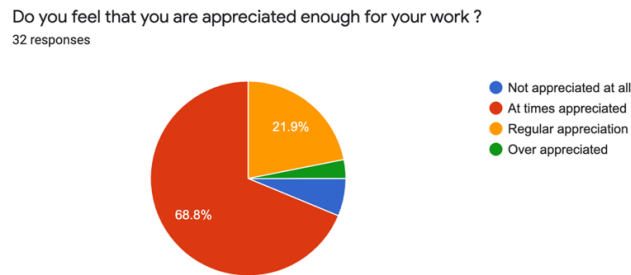


Figure 1: On the question of how often IU Dining employees are appreciated at work we found the above results.

for the food and awareness of the effort that is put in by the invisible workers.

2.1 Method

2.1.1 Survey. For our first assessment research, we recruited individuals working at IU (Indiana University) dining and students who eat at IU Dining. Initially, we rolled out a survey questionnaire with two groups of members- students working at IU dining and students who eat at those diners. There were around 13 questions of which some gathered demographic data for future interviews and the remaining questions related to gratitude sharing and workplace satisfaction. These surveys were designed and deployed on an online basis and collected data in which we got 32 responses. The survey was also a means to help recruit individuals for participation in our interview sessions later.

2.1.2 Interviews. We decided to conduct semi-structured interviews with some of the participants of the survey based on the responses they provided in the earlier survey to get more insights. We interviewed each member for around 20 to 30 minutes in a semi-structured way. In total we took 6 interviews, 3 with the IU Dining employees (invisible workers) and 3 students who visit IU Diners as customers. We wanted to understand whether these employees and service receivers had instances to interact with customers (service receivers) and IU Dining employees (invisible workers) respectively. We started the interview by making participants recall their experiences at IU Dining which then led to a detailed conversation about appreciation and recognition in terms of gratitude. There were 3 female and 3 male participants distributed during the interviews and we strived to recruit IU Dining employees working in distinct roles through the data we gathered in the earlier surveys.

2.2 Findings

The responses from the survey revealed that there are many IU Dining employees who are dissatisfied with their work and receive minimal appreciation for the work they did. One of the few key takeaways of the survey was when asked how much appreciation is received in general at the IU Dining job, 68.8% of the individuals thought they were appreciated at times, whereas 21.9% of the individuals felt they were appreciated regularly.

In another question whether these IU Dining employees have any interaction with the customers, it was surprising to know 56%

of the individuals said that they did have some sort of interaction with their customers. Through surveys, we were also able to gather data on the distinct roles played by the IU Dining employees. Of these, the roles varied like the roles of cashier, plating and server, cleaning, stocking, inventory management, server supervisor. This made us realize how the vast number of employees with varied roles make the IU Diner system work and there were even more roles that did not have much visibility or at times no visibility towards the customers. Another key finding was about the mode of appreciation that the employees would want during their work. Out of the 32 entries, 20 individuals decided to answer this question and around 50% of the individuals felt that some sort of gesture was essential to feel appreciated and 25% of the individuals felt they wanted an appreciation in the form of tips or monetary benefit [Fig 1]

It was a surprising finding when asked whether the IU Dining employees had any direct interaction with the customers, we found that 56.3% of the individuals had a brief amount of interaction with the customer while handing over the takeout meal boxes, but it was not enough to engage in meaningful conversations to show appreciation. These could be taken as “Missed chances” of showing gratitude as more people interacted with the employees [Fig 2]

We analyzed the data from surveys and interviews to look for pain points and moments to help us understand the scope of the problem and moments where gratitude can play a role in the overall problem space. we came up with three themes about the exchange of gratitude between the invisible workers and students at IU Diners

- (a) Gratitude as a key factor influencing their work
- (b) Visibility and Awareness issues
- (c) Positives for dining workers

2.2.1 Gratitude as a key factor influencing Dining workers. There are IU Dining employees who feel that they are not appreciated in a meaningful way for their work and that affects their enthusiasm for work. The dining workers usually have repetitive monotonous work unless the roles are changed for the very day, hence long hours of tiresome work with very few intermittent breaks are usually robotic and tedious. Hence appreciation of their work in the form of gestures, especially according to our survey, would help in motivating them to work with much more enthusiasm.

“Usually, my shift starts with a rush period, simultaneous orders are received which keeps me super busy with very few breaks”

Do you have any direct interactions with the customers regularly?
32 responses

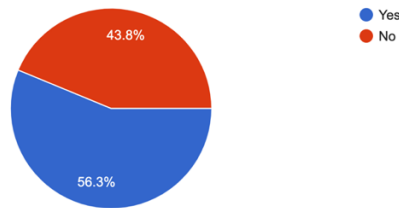


Figure 2: On the question of whether IU Dining employees had any direct interaction with customers, we found the above results.

-IU Dining Employee, Female, International

Our survey results supported this finding with 75% of the responses indicated that they had very few appreciations or did not receive appreciation at all.

2.2.2 Visibility and Awareness issues. The issue of visibility and awareness among the service receivers who were students in our case was also noticeably clear through our interviews. There was very minimal communication that could have taken place between the IU dining workers (invisible workers) and the students who were the service receivers.

“I don’t think that I have ever interacted with anyone besides the person at the counter and the server. Is that even possible? I mean most of them work in the kitchen and I haven’t really come across any situation that requires me to engage with them.”

-Graduate Student, Male, International

“Not many people think about the moving parts, the role we play unlike the frontend-staff, it’s not as recognized as it can be.”

-IU Dining Employee, Male, International

2.2.3 Positives for a Dining worker. There were quite a few positives that could be a defining factor for these employees. Especially through our interviews we could gather that many of the employees enjoy networking, exchanging experiences, bonding with their fellow dining workers. It was this community feeling and personalization of interaction which made them happy at their work.

“If something gets personalized it is better, or else it is seen as a courtesy and it is not that effective “

-IU Dining Employee, Male, International

“People from different backgrounds and are from different experiences, hence diversity and interaction help which motivates”

-IU Dining Employee, Male, International

You get to meet new people daily if at the front serving, their experiences and interaction is one main reason to be motivated effective, if a customer is given extra fries or extra products, they are very thankful “

-IU Dining Employee, Female, International

3 DESIGN IDEATION

Based on the identified major themes, each team member came up with design ideas. We then used the method of conceptual relays

to build on each other’s ideas. In this method one member of the team sketches his design idea on the whiteboard, then another team member builds on it, and this continues with every member of the team. After completing this method, we came up with a few micro design ideas and on further discussion we refined and combined them together. Then, we sketched high level scenarios and storyboards to visualize our design idea.

The design ideas were:

1. Enabling Kiosk to show the efforts that have gone for today’s meal, that can include approximate numbers of IU dining employees and staff, the amount of food, wastage of food from previous day.
2. A beacon notification triggered on IU student’s mobiles at IU dining that show them information about the labor that goes in to making the food. The notification will have a link through which students can share gratitude in terms of tips or even just a genuine message to an invisible worker.
3. Enabling a barcode-based kiosk that students can scan and share personalized appreciation message that will be visible at all IU dining. The idea behind this design was to invoke an emotion of gratitude among students by showing how others are sharing it.

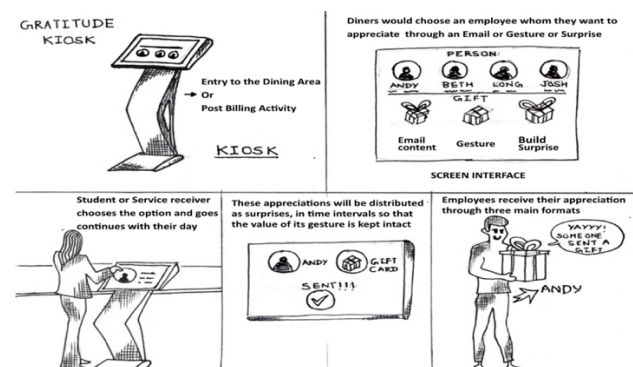


Figure 3: Initial sketch created through Conceptual Relays

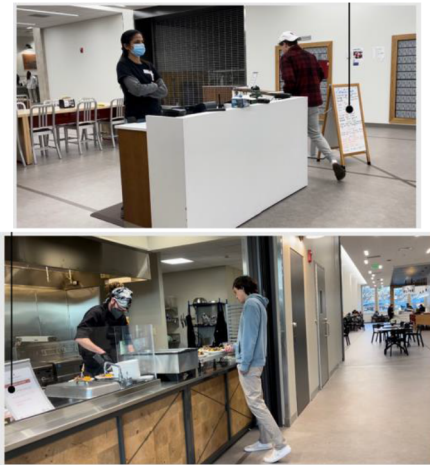


Figure 4: Dining Entry and Customers had very minimal waiting time around the proposed locations in our first ideation phase: Serving station

3.1 Ideating through first design iteration

Overall, all scenarios hover on a common idea to acknowledge and encourage an attitude of gratitude towards communities that work silently behind the scenes ‘invisible workers’.

Hence in our first iteration we tried to convey the message through a kiosk either at the entry or at the collection of their food stations.[Fig 3]

To create awareness and increase visibility we also came up with visual cues for the help service receivers in knowing who are working in that shift to help personalize their messages of appreciation. These boards were given anonymous names which corresponded to the actual names of the employees, hence keeping the names confidential but also making sure the heartfelt appreciation was sent to the employee.

3.2 Challenges through Initial Design

The main criteria for creating awareness and visibility were the interaction of customers with the kiosk or board of information, for our study we used a White Board to see the experience of the customers at two positions of the dining hall.

One was at the entry or the register of the dining hall and the second was a panel set up in the serving station of the dining hall.

These highlights were unable to create awareness and two main shortcomings were

1. Customers did not wait at the server station as their hands are usually full of plates and food hence interaction with any physical device was minimal unless it was cashier [Fig 4]

2. The entry point to dining areas were usually crowded hence people tried to go through without looking at the awareness points written on a board, or any information described around it.

Solution Components



Figure 5: Design Components for the GratiFood help service receivers communicate with invisible workers

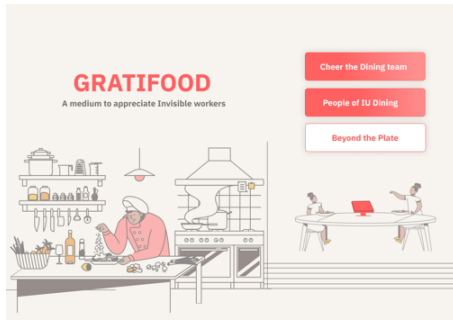


Figure 6: Proposed Home Screen user interface design

4 PROPOSED DESIGN

Hence iterating on the first design we decided to conceptualize the kiosk at the dining desk. The final design proposed is for a dining kiosk which will be named GratiFood, an interactive prototype that would encourage users to interact with the device and help share gratitude in the form of personalized messages, tips, or even an acknowledgment message.

GratiFood consists of a physical tabletop prototype kiosk with an iPad UI which would help users interact with the design. It consists of 3 modules to keep the interaction minimal but effective, with the diners (students in this case) to send their appreciation digitally [Fig 5].

These are designed with keeping in mind 3 aspects, mainly the dining teams-oriented appreciation, individual appreciation, and a module for creating awareness by showcasing interesting facts, history and information about food served at the dining area.

The Home Screen would act as landing screen to help diners explore the mode through which they can show their gesture of appreciation for the meals. [Fig 6].

The Interface screen prototypes are as follows:

The team appreciation helps the IU dining employee engaged in rush hours who are busy attending their customers and dining requests. These employees usually have their hands full and are engaged with stocking utensils, cleaning, plating food, coordinating with dining team mates hence a mode to reach out and send some gratitude would be through Voice or Audio. Hence after students fill up the Cheer-O-meter on the kiosk an appreciation would be sent to these teams at specific serving station. Since the timing of the sending these are depended on the cheer-o-meter getting filled there is a element of surprise, and would act as a genuine appreciation created by multiple diners to dining teams. [Fig 7]

Individual appreciation can be showcased through the people of IU Diners at the home screen and diners, or service receivers can show personal appreciation and add digital notes that would be sent to the respective dining employee.

Proposed way of sending this appreciation is either by email or when the employee checks in or checks out of the shift. [Fig 8] The screens may include few more development cycles at this point, as we would like to acknowledge for the partial completion of the design case. It would require further update and final tests with the students at the dining areas.



Figure 7: Proposed dining team appreciation screen design

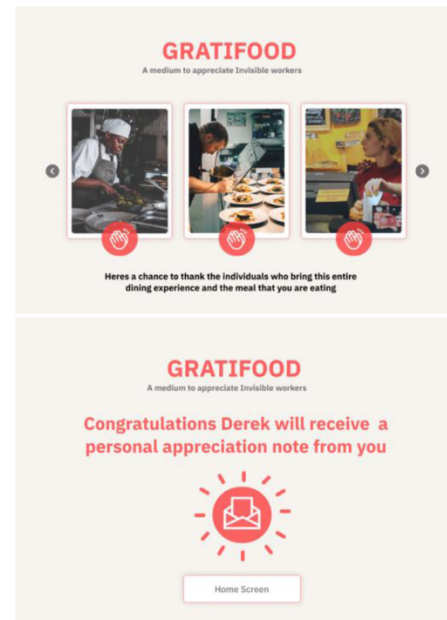


Figure 8: Proposed dining individual appreciation screen design

5 CONCLUSION

In this paper, we investigated the relation between gratitude and how monotonous work lifestyle of dining workers' demands appreciation from their service receivers. Through iterations of user research, we found that GratiFood explored technology would act as a catalyst to stimulate intervention which can lead to people acknowledging the effort and arouse gratitude for invisible workers and for food. The case of university dining would serve as a

steppingstone for a broader community of invisible workers and increase recognition and help build morale of long working hours for the dining workers. This would hence benefit all the individuals involved in this process find a wholesome experience in showing gratitude towards others.

REFERENCES

- [1] Ackerman E. Courtney. 2019. My Pocket Gratitude: Anytime Exercises for Awareness, Appreciation and Joy
- [2] McCullough, M. E. (2002). Savoring Life, Past and Present: Explaining What Hope and Gratitude Share in Common. *Psychological Inquiry*, 13(4), 302–304. <http://www.jstor.org/stable/1448873>
- [3] Kawasaki, Y., & Akamatsu, R. (2020). Appreciation for food, an important concept in mindful eating: association with home and school education, attitude, behavior, and health status in Japanese elementary school children. *Global health promotion*, 27(3), 140–149. <https://doi.org/10.1177/1757975919875650>
- [4] Martinec Novakova, Lenka & Bojanowski, Viola & Havlicek, Jan & Croy, Ilona. (2012). Differential Patterns of Food Appreciation during Consumption of a Simple Food in Congenitally Anosmic Individuals: An Explorative Study. *PloS one*. 7. e33921. 10.1371/journal.pone.0033921.
- [5] Leah R. Dickens (2017) Using Gratitude to Promote Positive Change: A Series of Meta-Analyses Investigating the Effectiveness of Gratitude Interventions, *Basic and Applied Social Psychology*, 39:4, 193–208, DOI: 10.1080/01973533.2017.1323638
- [6] Nowak, M. A., & Roch, S. (2007). Upstream reciprocity and the evolution of gratitude. *Proceedings. Biological sciences*, 274(1610), 605–609. <https://doi.org/10.1098/rspb.2006.0125>
- [7] Floyd Simeon, Rossi Giovanni, Baranova Julija, Dingemanse Mark, Kendrick Kobin H., Zinken Jörg and Enfield N. J. 2018 Universals and cultural diversity in the expression of gratitude *R. Soc. open sci.* 5180391 <http://doi.org/10.1098/rso.180391>
- [8] Bartlett, M. Y., & DeSteno, D. (2006). Gratitude and Prosocial Behavior: Helping When It Costs You. *Psychological Science*, 17(4), 319–325. <https://doi.org/10.1111/j.1467-9280.2006.01705.x>
- [9] Sansone, R. A., & Sansone, L. A. (2010). Gratitude and well being: the benefits of appreciation. *Psychiatry (Edgmont (Pa. : Township))*, 7(11), 18–22
- [10] Allen, Summer. 2008. The Science of Gratitude, Greater Good Science Center, White Paper prepared for John Templeton Foundation by Greater Good Science Center at UC Berkley